Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–VII (NEW) EXAMINATION – WINTER 2023

Subject Code:3174201 Date:01-12-2023

Subject Name: Deep Learning: Principles and Practices

Time: 10:30 AM TO 01:00 PM Total Marks:70

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			MARKS
Q.1	**(a)**	What are deep neural networks (DNNs)?	**03**
	(b)	How do neural networks mimic the functioning of the human brain?	**04**
	(c)	List and explain some of the major challenges in the field of deep learning.	**07**
Q.2	**(a)**	What is overfitting in the context of (Multilayer Perceptron) MLP? List some effective strategies to prevent overfitting.	**03**
	(b)	Justify the role of activation functions in Neural Network.	**04**
	(c)	What is an Artificial Neural Network (ANN)? How to calculate the output of a neuron in an ANN? Explain in detail.	**07**
		OR	
	(c)	What is the vanishing gradient problem? How does it affect training?	**07**
Q.3	**(a)**	What is a Neural Network Based Fuzzy System? What is its primary purpose in deep learning?	**03**
	(b)	How can Neural Network Based Fuzzy System (NN-FS) be integrated with deep learning architectures like convolutional neural networks or recurrent neural networks?	**04**
	(c)	Describe how neural networks can be used to realize basic fuzzy logic operators like AND, OR, and NOT.	**07**
		OR	
Q.3	**(a)**	Provide examples of recent real-world applications of NN-FS.	**03**
	(b)	Discuss the challenges and limitations of NN-FS.	**04**
	(c)	Explain the concept of neural network-based fuzzy logic inference. How does it work to make decisions based on fuzzy rules?	**07**
Q.4	**(a)**	What is rule-based neural fuzzy modeling?	**03**
	(b)	What is TensorFlow, and why is it widely used in deep learning?	**04**
	(c)	Discuss the trade-off between Gradient Descent and mini batch Gradient Descent in deep learning optimization.	**07**
		OR	
Q.4	**(a)**	What are some advantages of using neural networks to implement fuzzy logic operations?	**03**
	(b)	How does TensorFlow enable GPU and TPU acceleration for deep learning tasks?	**04**
	(c)	What is the Adam optimization algorithm, and how does it address some of the challenges of traditional gradient descent methods?	**07**

Q.5	**(a)**	Define CNN (Convolutional Neural Netwok). Enlist four applications of CNN.	**03**
	(b)	What is the key limitation of traditional RNN, and why does this limitation make them less suitable for capturing long-range dependencies?	**04**
	(c)	Provide a detailed explanation of how the training process works for both the Generator and Discriminator in a Generative Adversarial Network (GAN).	**07**

OR

Q.5	**(a)**	What does the forget gate in an LSTM unit do?	**03**
	(b)	Compare and contrast Radial Basis Function Network (RBFN) and MLP.	**04**
	(c)	Explain the Kohonen Self Organizing Map (KSOM) architecture, including its input layer, weight vectors, and clusters.	**07**

****